



Effect of baicalin on toll-like receptor 4-mediated ischemia/reperfusion inflammatory responses in alcoholic fatty liver condition

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ARTICLE INFO

Article history:

Received 1 September 2011

Revised 4 October 2011

Accepted 6 October 2011

Available online 14 October 2011

Keywords:

Alcoholic fatty liver

Baicalin

Inflammation

Ischemia/reperfusion

Toll-like receptor

ABSTRACT

Alcoholic fatty liver is susceptible to secondary stresses such as ischemia/reperfusion (I/R). Baicalin is an active component extracted from *Scutellaria baicalensis*, which is widely used in herbal preparations for treatment of hepatic diseases and inflammatory disorders. This study evaluated the potential beneficial effect of baicalin on I/R injury in alcoholic fatty liver. Rats were fed an alcohol liquid diet or a control isocaloric diet for 5 weeks, and then subjected to 60 min of hepatic ischemia and 5 h of reperfusion. Baicalin (200 mg/kg) was intraperitoneally administered 24 and 1 h before ischemia. After reperfusion, baicalin attenuated the increases in serum alanine aminotransferase activity, tumor necrosis factor- α (TNF- α), and interleukin-6 (IL-6) levels in alcoholic fatty liver. The increased levels of TNF- α and IL-6 mRNA expression and inducible nitric oxide synthase and cyclooxygenase-2 protein and mRNA expressions increased after reperfusion, which were higher in ethanol-fed animals, were attenuated by baicalin. In ethanol-fed animals, baicalin attenuated the increases in toll-like receptor 4 (TLR4) and myeloid differentiation factor 88 protein expressions and the nuclear translocation of NF- κ B after reperfusion. In conclusion, our findings suggest that baicalin ameliorates I/R-induced hepatocellular damage by suppressing TLR4-mediated inflammatory responses in alcoholic fatty liver.

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Introduction

Hepatic ischemia/reperfusion (I/R) is a common problem following many clinical conditions, such as liver surgery, liver transplantation, trauma and cancer. Liver transplantation is an increasingly accepted intervention for end-stage liver disease. However, many patients on waiting lists die each year due to a shortage of liver donors. For this reason, many researchers are attempting to maximize the usage of marginal livers (Adair et al., 2011). Fatty liver disease, primarily caused by chronic ethanol consumption and/or obesity, is the most prevalent underlying condition affecting human liver donors and presents a particular challenge in liver transplantation (Szczepaniak et al., 2005). In particular, alcoholic fatty liver in humans is susceptible to greater I/R injury and a higher incidence of primary graft non-function after transplantation (Selzner and Clavien, 2001).

Although the exact molecular mechanisms of hepatic I/R injury remain evasive, previous reports suggest that the pathophysiology of hepatic I/R is characterized by a considerable inflammatory response as well as liver cell death, which results in graft organ failure (Fondevila

et al., 2003). Prolonged ischemia leads to cell necrosis, which causes the release of a variety of pro-inflammatory mediators including cytokines and reactive oxygen and nitrogen radicals. After the restoration of blood supply, reperfusion not only promotes an inflammatory response, but also evokes hepatocellular apoptosis (Cursio et al., 2010). Moreover, the importance of inflammatory pathways caused by chronic ethanol intake is well known (Mandal et al., 2010).

Toll-like receptors (TLRs) are a family of proteins that recognize microbial components, pathogen-associated molecular patterns (PAMPs). Recent studies have revealed that TLR-mediated signals are involved in endogenous danger-associated molecular patterns (DAMPs) including I/R liver injury (Seki and Brenner, 2008). In particular, hepatic I/R injury is exacerbated by activation of TLR4 by high-mobility group box1 (HMGB1), a DAMP protein released from dying cells. Myeloid differentiation factor 88 (MyD88)-dependent signals in the TLR4 activation pathway lead to direct nuclear factor- κ B (NF- κ B) activation and subsequent induction of pro-inflammatory cytokines (Avbelj et al., 2011; Tsung et al., 2005). Additionally, recent studies have reported the involvement of TLRs in the early stages of alcoholic liver diseases (French et al., 2010).

Scutellaria baicalensis (Lamiaceae) is an eastern traditional medicine that is used as an anti-inflammatory agent and as a component of some hepatoprotective herb mixtures in China, Japan, and Korea (Taira et al., 2004). Baicalin is the major active constituent of the isolated root of *S. baicalensis*; baicalin scavenges reactive oxygen radicals and exerts an anti-inflammatory activity (Li et al., 2000). According to recent reports, baicalin plays a beneficial role in various experimental

Abbreviations: I/R, ischemia/reperfusion; TLR, toll-like receptor; PAMP, pathogen-associated molecular pattern; DAMP, danger-associated molecular pattern; HMGB1, high-mobility group box1; MyD88, myeloid differentiation factor 88; NF- κ B, nuclear factor- κ B; CD, control diet; ED, ethanol diet; TG, triglyceride; TNF- α , tumor necrosis factor- α ; IL-6, interleukin-6; iNOS, inducible nitric oxide synthase; COX-2, cyclooxygenase-2.

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models of invasive toxicants by alleviating inflammatory responses (Lixuan et al., 2010; Park et al., 2008; Qiao et al., 2010). We also reported the cytoprotective effect of baicalin against I/R injury in normal rat liver (Kim et al., 2010). Guo et al. (2009) demonstrated that baicalin treatment protected against the development of hepatic steatosis and obesity induced by a long term high-fat diet via enhancement of hepatic AMPK activation. However, in models of alcoholic fatty liver, there is a shortage of information about the beneficial effects of baicalin against hepatic I/R injury.

Therefore, we have employed the Lieber-DeCarli chronic ethanol consumption model to mimic human alcoholic fatty liver disease, as this model may closely resemble its clinical status (de la M Hall et al., 2001; Lieber and DeCarli, 1989). The aims of this study were to examine the effect of baicalin against I/R insult in alcoholic fatty liver and to investigate its possible mechanisms of action on inflammatory factors.

Materials and methods

Animals. Male Sprague–Dawley rats weighing 150–170 g were acclimatized to laboratory conditions at Sungkyunkwan University for at least one week. Animals were kept in a temperature- and humidity-controlled room ($25 \pm 1^\circ\text{C}$ and $55 \pm 5\%$, respectively) under a 12-h light–dark cycle. This study was approved by the Animal Care Committee of Sungkyunkwan University, Korea, and all animals were handled according to the Guide for the Care and Use of Laboratory Animals published by the National Institutes of Health.

Liquid diet technique for induction of alcoholic fatty liver. Due to the tendency of animals given alcohol to reduce their solid food consumption, animals were given a liquid diet. The rats were randomly assigned to the groups specified and given free access to Lieber-DeCarli liquid diet from Dyets Inc. (Bethlehem, PA); 36% of the total calories were supplied from ethanol (ethanol group) or maltodextran (control group) (Lieber and DeCarli, 1989). Ethanol was introduced progressively at 30 g/l of liquid diet for 2 days, 40 g/l for the two following days, and 50 g/l thereafter (Kerai et al., 1998). Animal body weights and general condition were monitored twice weekly, and liquid diet intake was determined daily. Animals were assigned randomly to receive a control diet for 4 weeks or an ethanol isocaloric liquid diet for the same period. Ethanol was incorporated into the liquid feed immediately before feeding.

Hepatic ischemia procedure. Rats were anesthetized with an intraperitoneal injection of xylazine (8 mg/kg) and ketamine (60 mg/kg). A midline incision was made to the abdomen, and the left parts of portal vein, hepatic artery, and bile duct were clamped to induce complete ischemia of the median and left hepatic lobes. The right lobes remained perfused to prevent intestinal congestion. After 60 min of ischemia, the clip around the left branches of the portal vein was removed. Sham animals were prepared in a similar manner except that no clip was placed on the left and median lobes. Rats were sacrificed after 5 h of reperfusion, and blood samples were obtained from the abdominal aorta for the determination of serum alanine aminotransferase (ALT) activity. The livers were then excised and weighed immediately; samples from the left lobe were analyzed for any histological changes. The remaining major portions of the liver tissues and serum were immediately stored at -75°C until biochemical analysis.

Drug administration and experimental groups. Baicalin (95% purity; Sigma Chemical Co., St. Louis, MO), dissolved in 10% Tween 80 (vehicle), was administered intraperitoneally (200 mg/kg) 24 h and 1 h before the sustained ischemia. The dose was selected based on our previous studies on its protective effect against hepatic I/R injury (Kim et al., 2010). The rats were divided into eight experimental groups in the study: (a) vehicle-treated sham, control diet (CD sham),

(b) baicalin-treated sham, control diet (CD baicalin), (c) vehicle-treated ischemic, control diet (CD I/R), (d) baicalin-treated ischemic, control diet (CD I/R + baicalin), (e) vehicle-treated sham, ethanol diet (ED sham), (f) baicalin-treated sham, ethanol diet (ED baicalin), (g) vehicle-treated ischemic, ethanol diet (ED I/R), and (h) baicalin-treated ischemic, ethanol diet (ED I/R + baicalin). Since there were no differences in any of the parameters between vehicle-treated sham and baicalin-treated sham groups, (a) and (b), (e) and (f) were pooled and are referred to as CD sham and ED sham, respectively.

Liver weight, serum triglyceride concentration and hepatic triglyceride content. To examine the extent of fatty livers, whole liver weights were measured 5 h after reperfusion and expressed as liver weight/100 g of body weight. Serum triglyceride (TG) concentrations were determined spectrophotometrically with a commercial kit (IVDLab Co. Ltd., Uiwang, Korea) using triglyceride (200 mg/dl) as the standard. To analyze hepatic TG content, livers were homogenized in buffer containing 0.25 M sucrose, 50 mM Tris–HCl and 1 mM EDTA (pH 7.4). Lipids were then extracted from the liver homogenates using a chloroform:methanol (2:1) mix and quantified as described by Folch et al. (1957).

Serum ALT activity. The serum ALT activity was determined by standard spectrophotometric procedures using a ChemiLab ALT assay kit (IVDLab Co., Ltd.).

Histological analysis. Liver tissues were removed from a portion of the left lobe, fixed immediately in 10% neutral buffered formalin, embedded in paraffin, and cut serially into 5 μm sections. The hematoxylin and eosin (H&E)-stained sections were evaluated using an optical microscope (Olympus Optical Co., Tokyo, Japan).

Serum tumor necrosis factor- α (TNF- α) and interleukin-6 (IL-6) levels. TNF- α and IL-6 levels were quantified using commercial TNF- α and IL-6 ELISA kits (eBiosciences Co., San Diego, CA), respectively.

Western blot immunoassay. Nuclear and cytosolic proteins were isolated from the fresh rat liver using NE-PER® Nuclear and Cytoplasmic Extraction Reagents according to the manufacturer's instructions (Pierce Biotechnology, Rockford, IL). 20 μg of soluble protein from liver homogenates was used to determine the content of TLR4, MyD88, inducible nitric oxide synthase (iNOS) and cyclooxygenase-2 (COX-2), and 20 μg of nuclear protein was used to determine the content of NF- κB p65. 20 μg of cytosolic protein was used to determine I κB - α expression. Protein samples were mixed with 2 $\times$ reducing buffer [125 mM Tris–HCl, pH 6.8, 20% glycerol, 4% sodium dodecyl sulfate (SDS), 10% 2-mercaptoethanol and 0.02% bromophenol blue], sonicated for 5 s, and heated for 7 min in boiling water. Denatured proteins were separated by 10% SDS-PAGE and were transferred to nitrocellulose membranes; the membranes were then incubated at room temperature for 60 min in buffer containing 5% dried skim milk to block the non-specific binding. Bands were immunologically detected using polyclonal antibodies against rat TLR4 (Santa Cruz Biotechnology, Santa Cruz, CA; 1:2500), MyD88 (Santa Cruz Biotechnology; 1:2000), iNOS (Cell Signaling Technology, Irvine, CA; 1:2000), COX-2 (Cayman Chemical Company, Ann Arbor, MI; 1:2000), NF- κB p65 (Santa Cruz Biotechnology; 1:1000), I κB - α (Santa Cruz Biotechnology; 1:1000), lamin B1 (Abcam, Cambridge, UK; 1:2500) and monoclonal antibody against rat β -actin (Sigma Chemical Co.; 1:5000). The binding of all the antibodies was detected by an ECL detection system (iNtRON Biotechnology Co. Ltd., Seongnam, Korea) according to the manufacturer's instructions. The visualized immunoreactive bands were evaluated densitometrically with ImageQuant™ TL software (Amersham Biosciences/GE Healthcare, Piscataway, NJ). The signals were normalized to that of β -actin for whole liver and cytosolic proteins or lamin B1 for nuclear proteins.

Reverse Transcription Polymerase Chain Reaction (RT-PCR). Reverse transcription of the total RNA extracted from the tissue samples was carried out in order to synthesize first-strand cDNA using the oligo (dT) 12–18 primer and Takara RNA PCR™ AMV Kit (Takara Bio Inc., Shiga, Japan). The PCR reaction was performed with a diluted cDNA sample in a 20 µl reaction volume. The final reaction concentrations were as follows: primers, 10 µM; dNTP mix, 250 µM; 10× PCR buffer; and Ex Taq DNA polymerase, 0.5 U per reaction. The gene-specific primers are listed in Table 1. RT-PCR was carried out with an initial denaturation step at 94 °C for 5 min and a final extension step at 72 °C for 7 min using GeneAmp 2700 thermocycler (Applied Biosystems, Foster City, CA). The amplification cycling conditions were as follows: 30 cycles of 30 s at 94 °C, 30 s at 56 °C and 60 s at 72 °C for *TNF-α*; 40 cycles of 30 s at 94 °C, 45 s at 65 °C and 45 s at 72 °C for *IL-6*; 32 cycles of 45 s at 94 °C, 45 s at 65 °C and 60 s at 72 °C for *iNOS*; 40 cycles of 30 s at 94 °C, 30 s at 65 °C and 60 s at 72 °C for *COX-2*; 25 cycles of 30 s at 94 °C, 30 s at 62 °C and 60 s at 72 °C for *β-actin*. After RT-PCR, 10 µl samples of the PCR products were visualized by ultraviolet illumination after electrophoresis through 1.5% agarose gel and ethidium bromide staining. The intensity of each PCR product was analyzed semiquantitatively using a digital camera (DC120, Eastman Kodak, New Haven, CT) and analysis software.

Statistical analysis. All results are reported as the means ± SEM. The overall significance of the data was examined by two-way analysis of variance (ANOVA). The differences between the groups were considered significant at $P < 0.05$ with the appropriate Bonferroni correction made for multiple comparisons.

Results

Induction of alcoholic fatty liver

The liver weight to body weight ratios after feeding Lieber-DeCarli diet for 5 weeks are presented in Table 2. Liver weight to body weight ratio in the animals fed with control diet was 2.8 ± 0.1 g liver/100 g body weight; the liver weight to body weight ratio in animals fed with ethanol diet (3.3 ± 0.3 g liver/100 g body weight) was higher than that in animals fed with control diet. TG content from liver tissue and serum in animals fed with control diet for 5 weeks were 11.3 ± 1.12 mg/g liver and 19.7 ± 1.75 mg/dl, respectively. In animals fed with the ethanol diet, hepatic TG content significantly increased to 21.6 ± 1.09 mg/g liver, and serum TG concentration increased to 39.7 ± 7.11 mg/dl (Table 2).

Serum alanine aminotransferase activity and histological analysis

As shown in Table 3, the serum ALT activity was 52.3 ± 12.9 U/l in the CD sham group. After reperfusion, the serum ALT activity increased significantly to 3899.4 ± 6.7 U/l, and baicalin treatment significantly attenuated this increase. The serum ALT activity was 59.7 ± 6.7 U/l in the ED sham and it was not different in the CD

Table 2

Effect of chronic ethanol consumption on liver weight, serum triglyceride concentration and liver triglyceride content.

	Liver weight (g liver/100 g body weight ⁻¹)	Liver TG content (mg/g liver)	Serum TG concentration (mg/dl)
CD	2.8 ± 0.1	11.3 ± 1.1	19.7 ± 1.8
ED	$3.3 \pm 0.3^{**}$	$21.6 \pm 1.1^{**}$	$39.7 \pm 7.1^{**}$

Each value is the mean ± SEM of 8–10 rats per group. CD, control diet; ED, ethanol diet; TG, triglyceride.

** Significantly different ($P < 0.01$) from CD.

sham. After reperfusion, serum ALT activity in the ED I/R (5573.2 ± 620.4 U/l) increased significantly compared with the ED sham, which was more remarkable than that of the CD I/R. This increase was attenuated by baicalin. As shown in Fig. 1, hepatocytes were well arranged and had normal morphology in the CD sham group. Mixed micro- and macrovesicular steatosis scattered in the liver lobule was seen in the ethanol-fed group, confirming the reported mild fatty infiltration in the Lieber-DeCarli model compared with other models of chronic alcohol feeding. After reperfusion, excessive hemorrhagic necrosis, extensive areas of portal inflammation, and increase in inflammatory cell infiltration with damaged tissue structure were observed both in the CD I/R and ED I/R group, and these changes were more remarkable in the ED I/R group. However, these changes were attenuated in baicalin-pretreated animals.

Hepatic TLR4 and MyD88 protein expression

Although the hepatic levels of TLR4 and MyD88 protein expression of ED sham increased compared to those of CD sham, there were no statistically significant differences between CD sham and ED sham. The hepatic levels of TLR4 and MyD88 protein expression significantly increased after reperfusion in CD animals (3.6-fold and 13.1-fold vs. CD sham, respectively). In animals fed the ethanol diet, the levels of TLR4 and MyD88 protein expression increased after reperfusion (3.2-fold and 14.7-fold vs. ED sham, respectively), and the increased levels were significantly higher than those in CD I/R (1.3-fold and 1.8-fold, respectively). However, the increased levels of TLR4 and MyD88 protein expression were attenuated by baicalin (Fig. 2).

IκB-α degradation, NF-κB/p65 translocation

There were no significant differences in levels of nuclear NF-κB/p65 and cytosolic IκB-α protein expression between CD sham and ED sham. After reperfusion, the level of nuclear NF-κB/p65 protein expression markedly increased by 2.6-fold in the animals fed the control diet. In animals fed an ethanol diet, the level of nuclear NF-κB/p65 protein expression increased significantly by 3.2-fold after

Table 1
RT-PCR primers used in study.

Gene	Accession number	Primer sequences (5' → 3')	Product length (bp)
<i>TNF-α</i>	NM_012675	Sense: GTA GCC CAC GTC GTA GCA AA Anti-sense: CCC TTC TCC AGC TGG AAG AC	347
<i>IL-6</i>	NM_012589	Sense: GAA AGT CAA CTC CAT CTG CC Anti-sense: CAT AGC ACA CTA CGT TTG CC	679
<i>iNOS</i>	NM_012611	Sense: TTC TTT GCT TCT GTG CTT AAT GCG Anti-sense: GTT GTT GCT GAA CTT CCA ATC GT	3793
<i>COX-2</i>	NM_017232	Sense: CTG CAT GTG GCT GAT GTC ATC Anti-sense: AGG ACC CGT CAT CTC CAG GGT AAT C	474
<i>β-actin</i>	NM_031144	Sense: TTG TAA CCA ACT GGG ACG ATA TGG Anti-sense: GAT CTT GAT CTT CAT GGT GCT AG	764

Table 3

Effect of Baicalin on Serum ALT Activity after Hepatic Ischemia/Reperfusion in Alcoholic Fatty Liver.

	ALT (U/l)
CD	
Sham	52.3 ± 12.9
I/R	3899.4 ± 351.6**
I/R + baicalin	2115.2 ± 193.6##
ED	
Sham	59.6 ± 6.7
I/R	5573.2 ± 620.4**,#
I/R + baicalin	1529.0 ± 169.2††

Each value is the mean ± SEM of 8–10 rats per group. **Significantly different ($P < 0.01$) from each sham, group. ##Significantly different ($P < 0.05$, $P < 0.01$) from CD I/R.

††Significantly different ($P < 0.01$) from ED I/R. CD, control diet; ED, ethanol diet.

reperfusion, and this level was significantly higher than that in the CD I/R group (1.5-fold vs. CD I/R). In the CD and ED animals, the levels of cytosolic I κ B- α protein expression decreased significantly by 44.5% and 35.3% after reperfusion, respectively. However, baicalin attenuated the increase of nuclear NF- κ B/p65 protein expression and the decrease of cytosolic I κ B- α (Figs. 3A and B).

Serum TNF- α and IL-6 levels

There were no significant differences in serum TNF- α and IL-6 levels between CD sham and ED sham. The serum levels of TNF- α and IL-6 in the CD I/R group were approximately 2.7-fold and 2.2-fold higher than that of CD sham, respectively. After reperfusion, the serum level of TNF- α and IL-6 in the ED I/R group increased significantly by about 9.7-fold and 4.8-fold over the ED sham, respectively. In the ED I/R group, the increased levels of serum TNF- α and IL-6 were significantly higher than those in CD I/R (2.4-fold and 1.6-fold vs. CD I/R, respectively). However, baicalin treatment attenuated these increases (Figs. 4A and B).

Hepatic TNF- α and IL-6 mRNA expression

Although the hepatic levels of TNF- α and IL-6 mRNA expression in ED sham increased compared to those in CD sham, there were no statistically significant differences between CD sham and ED sham. After reperfusion, the levels of hepatic TNF- α and IL-6 mRNA expression significantly increased by 2.7-fold and 3.6-fold compared with those of CD sham, respectively. The hepatic levels of TNF- α and

IL-6 mRNA expression significantly increased after reperfusion (2.8-fold and 3.9-fold vs. ED sham) and were significantly higher than those of the CD I/R group. However, baicalin attenuated these increases (Figs. 4C and D).

Hepatic iNOS, COX-2 protein and mRNA expression

As shown in Figs. 5A and B, there were no significant differences in the levels of hepatic iNOS and COX-2 protein and mRNA expressions between CD sham and ED sham. The levels of iNOS protein and mRNA expression increased by 6.7-fold and 1.9-fold after reperfusion in CD animals, respectively. In animals fed the ethanol diet, the levels of iNOS protein and mRNA expression increased by 7.9-fold and 3.3-fold after reperfusion, respectively. Although the hepatic levels of COX-2 protein in ED sham increased compared to those in CD sham, there were no statistically significant differences between CD sham and ED sham. The levels of COX-2 protein and mRNA expression increased by 6.6-fold and 2.2-fold after reperfusion in animals fed the control diet, respectively. In animals fed the ethanol diet, the levels of COX-2 protein and mRNA expression increased by 5.2-fold and 2.8-fold after reperfusion, respectively. However, baicalin treatment attenuated these increases (Figs. 5C and D).

Discussion

Alcoholic fatty liver is a rapidly increasing risk factor for liver transplantation and liver resection, and for graft malfunction. Previous studies have reported that alcoholic fatty liver is vulnerable to I/R injury, and it might be due to excessive activation of inflammatory responses (Cho and Lee, 2007). Hepatic I/R injury may also be more intense in marginal livers such as fatty liver and aged liver, which comprise a large portion of possible donation livers. Alleviating the severity of marginal liver I/R injury and the subsequent complications is being urgently investigated for prospective remedies. We induced alcoholic fatty liver by the isocaloric feeding formula using the Lieber-DeCarli liquid diet, which induces mild hepatic injury that closely mimics the early stages of human alcoholic liver diseases. This animal model stimulates experimentally ethanol-selective phenomena that are distinguishable from those commonly associated with fatty liver. The exact mechanisms of I/R injury in the alcoholic fatty liver remain elusive, but are distinctive from those of non-alcoholic fatty liver.

During ischemia, the metabolic and respiratory processes required for energy production are severely compromised, resulting in lower cellular ATP levels. This is believed to be responsible for the necrosis

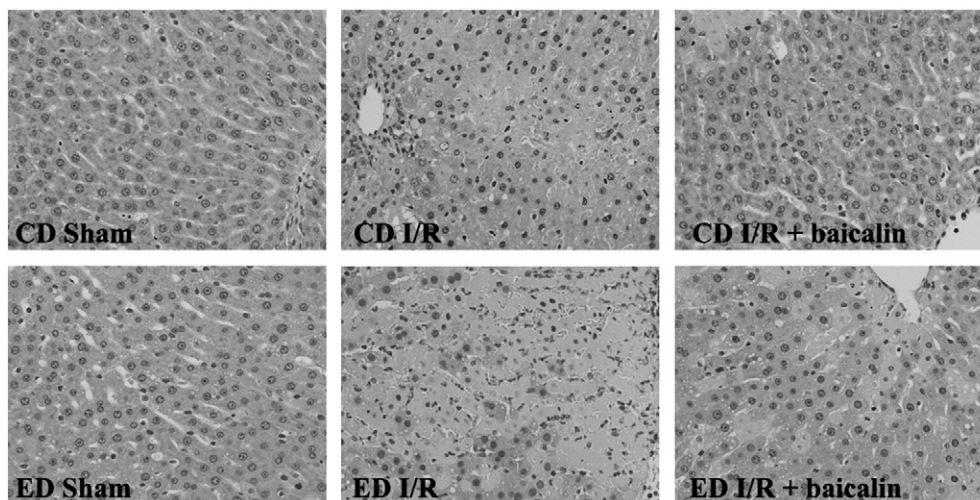


Fig. 1. Effect of baicalin on the histological changes of alcoholic fatty liver after hepatic I/R (original magnification 100 $\times$).

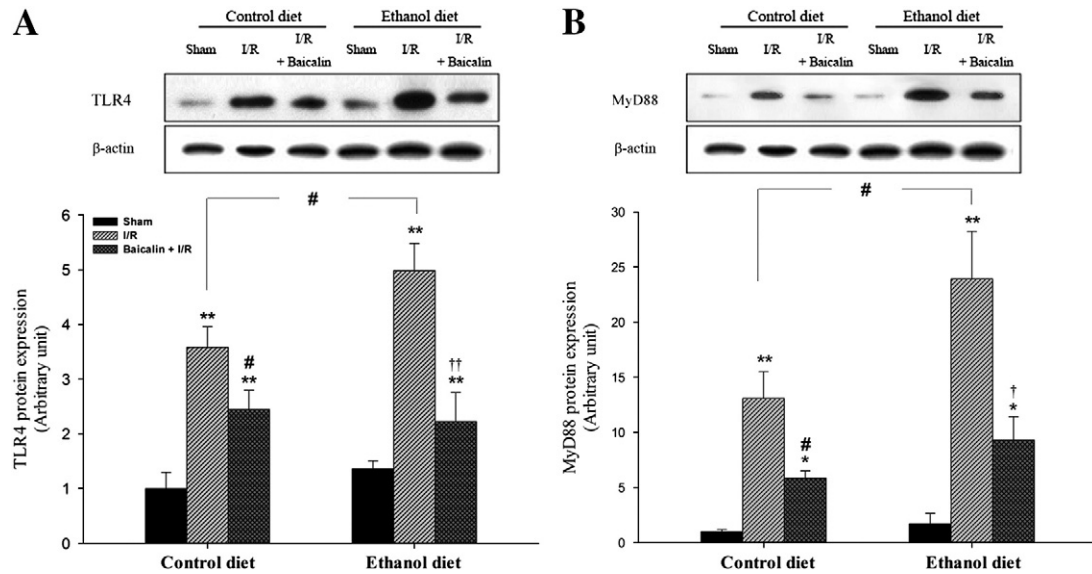


Fig. 2. Effect of baicalin on TLR4 and MyD88 protein expression after I/R in alcoholic fatty liver. A: hepatic TLR4 protein expression. B: hepatic MyD88 protein expression. The density values were normalized to β -actin. Values are means $\pm$ SEM for 8–10 rats per group. **Significantly different ($P < 0.05$, $P < 0.01$) from each sham group. #Significantly different ($P < 0.05$) from CD I/R. ††Significantly different ($P < 0.05$, $P < 0.01$) from ED I/R.

observed in liver tissues under I/R. Additionally, alcoholic fatty liver is associated with decreased ATP production and a disturbance of sinusoidal flow, which can cause a shortage of energy supply in peripheral liver tissue (Selzner and Clavien, 2001). In our study, a 60-min period of ischemia was used to explore the priming effects of ethanol on subsequent ischemic stress. In ischemic rats, the ALT activity increased markedly after reperfusion, and chronic ethanol consumption potentiated I/R-induced hepatic injury. A protective effect against I/R injury was evident in baicalin-treated animals, which suggests that baicalin offers protection by its antioxidant activity in reducing generation of reactive oxygen species and hence prevents hepatocellular necrosis against I/R injury in alcoholic fatty liver. This was supported by the histological findings showing hemorrhagic necrosis, portal inflammation and inflammatory cell infiltration with damaged tissue structure after reperfusion, and restoration in baicalin-treated animals.

Activation of macrophages plays a central role in the pathogenesis of I/R injury, particularly in the experimental model of alcoholic fatty liver (Yamada et al., 2003). TLRs are a class of pattern recognition receptor proteins and are expressed on Kupffer cells as well as on other cell types in the liver (Wu et al., 2010). In rodents, the TLR pathway takes a prime place in the pathogenesis of early ethanol-induced liver injury (Inokuchi et al., 2011). Enteral ethanol exposure increased the expression of TLR1, 2, 4, 6, 7, 8 and 9 liver mRNA (Gustot et al., 2006). Previous studies demonstrated a role for some subfamilies of TLRs during I/R-induced liver injury, including TLR2, TLR4 and TLR9 (Bamboate et al., 2010; Zhai et al., 2008; Zhang et al., 2004). In addition, Ellett et al. (2009) indicated the involvement of TLR4 in I/R-induced injury in non-alcoholic steatotic murine livers. In our study, I/R significantly increased the TLR4 and MyD88 protein expression levels, but failed to induce the protein expression of TLR2 and TLR9 in

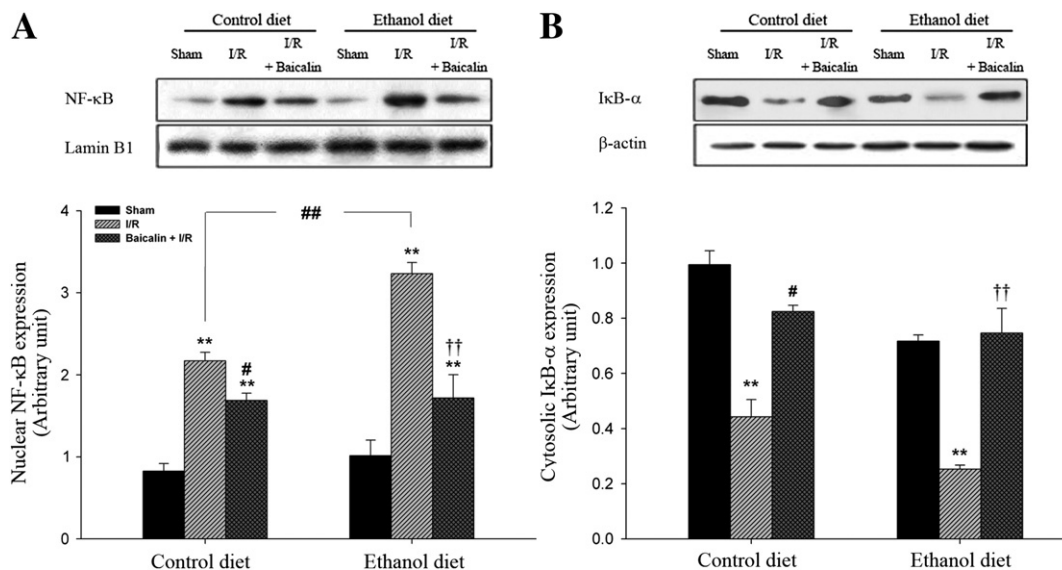


Fig. 3. Effect of baicalin on NF- κ B and I κ B- α protein expression after I/R in alcoholic fatty liver. A: nuclear NF- κ B/p65 protein expression. B: cytosolic I κ B- α expression. Density values of nuclear NF- κ B/p65 and cytosolic I κ B- α were normalized to nuclear lamin B1 and cytosolic β -actin, respectively. Values are means $\pm$ SEM for 8–10 rats per group. **Significantly different ($P < 0.01$) vs. each sham group. ###Significantly different ($P < 0.05$, $P < 0.01$) vs. CD I/R. ††Significantly different ($P < 0.01$) vs. ED I/R.

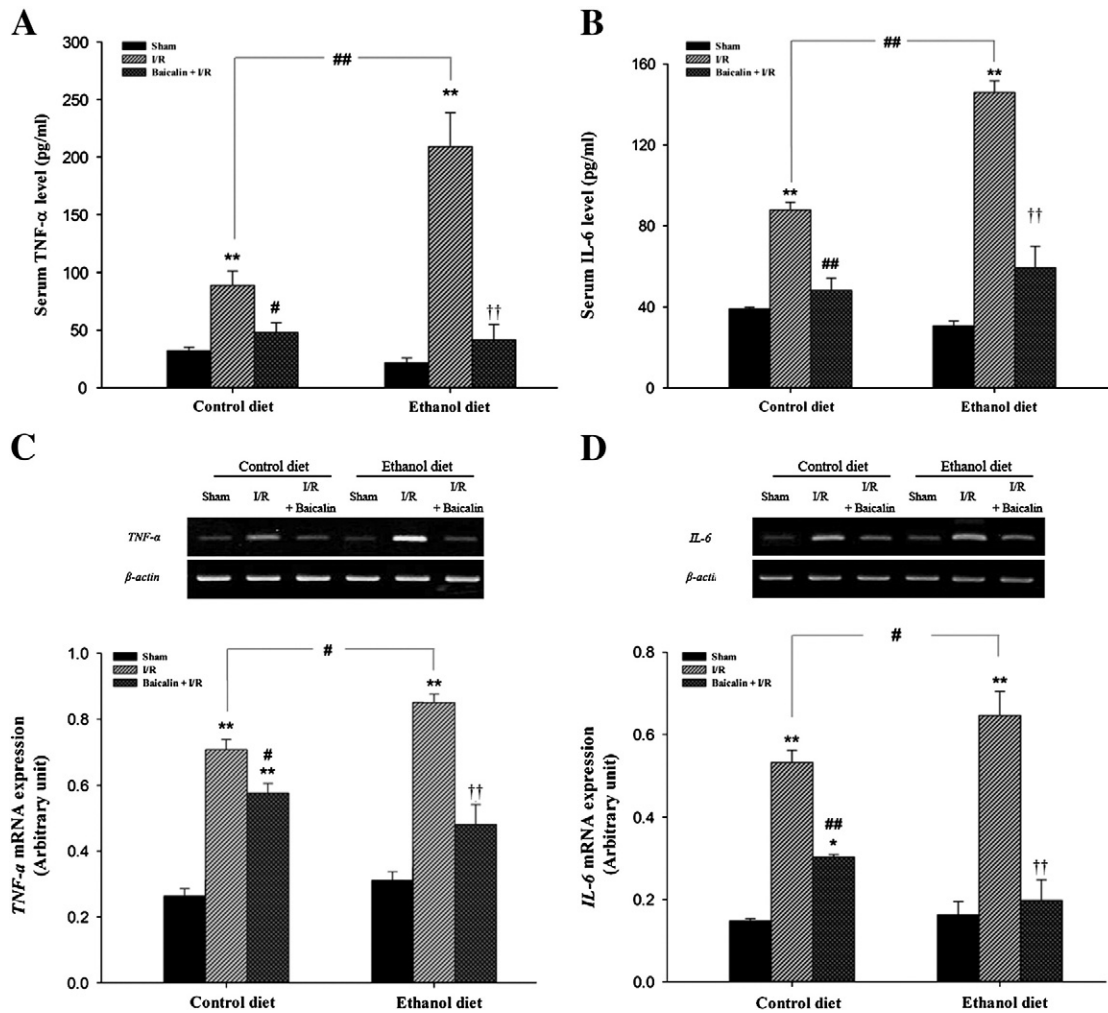


Fig. 4. Effect of baicalin on serum TNF- α , IL-6 levels and their mRNA expressions after I/R in alcoholic fatty liver. A: serum TNF- α level. B: serum IL-6 level. C: hepatic TNF- α mRNA expression. D: hepatic IL-6 mRNA expression. The density values were normalized to β -actin. Values are means $\pm$ SEM for 8–10 rats per group. **, ***Significantly different ($P < 0.05$, $P < 0.01$) vs. each sham group. ##, ###Significantly different ($P < 0.05$, $P < 0.01$) vs. CD I/R. ††Significantly different ($P < 0.01$) vs. ED I/R.

both CD and ED animals (data not shown). Chronic ethanol consumption alone slightly increased TLR4 and MyD88 expression. Moreover, the combination of both stresses had a synergistic effect. However, baicalin pretreatment reduced these changes.

Previous studies have demonstrated involvement of NF- κ B activation in TLR4-mediated hepatic I/R injury (Tsung et al., 2005). Furthermore, long-term ethanol intake potentiates the production of reactive oxygen species in the liver after I/R, which may enhance NF- κ B binding activity (Yamada et al., 2000). In our study, slightly elevated nuclear NF- κ B/p65 level was observed in ED animals compared to CD animals; these increases were further augmented after I/R in ethanol-fed animals, while baicalin reduced these alternations. These results suggest that baicalin exerts its protective effect by reducing the expression of TLR4 and its transmembrane signal transducers, MyD88 and NF- κ B, which play a critical role in the development of the I/R-induced inflammatory response in alcoholic fatty livers.

TNF- α is one of the most important factors responsible for I/R injury. Previous results showed that TLR4 expression is involved in the increased level of TNF- α during I/R (Kaczorowski et al., 2007; Kang et al., 2011). Furthermore, TNF- α appears to play a critical role in the development of steatosis in obesity- and ethanol-induced fatty livers (Hill et al., 2000). IL-6 is implicated in the development of the I/R injury and is correlated with the severity of I/R injury in a TLR4-dependent manner (Langdale et al., 2008; Uchida et al., 2010). IL-6 is elevated in the plasma and peripheral blood monocytes of patients

with fatty diseases, including alcoholic liver disease and non-alcoholic steatohepatitis (Hill et al., 1992). The circulating levels and hepatic gene expression of TNF- α and IL-6 were increased after I/R alone, but not after chronic alcohol consumption. Combining both stresses led to further synergistic induction, which would correspond with increased inflammatory response. However, baicalin pretreatment alleviated these changes.

NO is involved in acute liver injury based on observations that toxin-induced hepatic damage is associated with increased NO production by the liver (Li and Billiar, 1999). Although it remains to be determined whether the augmented production of NO has a protective or deleterious role in the liver, the detrimental effect of excessive NO produced by iNOS is well known (Yanagida et al., 2006). Imbalanced regulation of NO homeostasis caused by I/R could interfere the sinusoidal microcirculatory functions, which in turn detract from the hepatocellular function along with the other inflammatory cytokines and vascular mediators (Sonin et al., 1999). The involvement of TLR4 in the I/R-induced increase of iNOS was previously reported (Kaczorowski et al., 2007), and enhancement of iNOS gene production was observed in steatotic liver followed by normothermic ischemia (Koeppl et al., 2007).

COX-2, the inducible isoform of cyclooxygenase, is capable of producing large amounts of prostaglandins and is of particular interest due to its potential role in I/R injury (Hamada et al., 2008). Here we found that hepatic iNOS, COX-2 protein and gene expression were

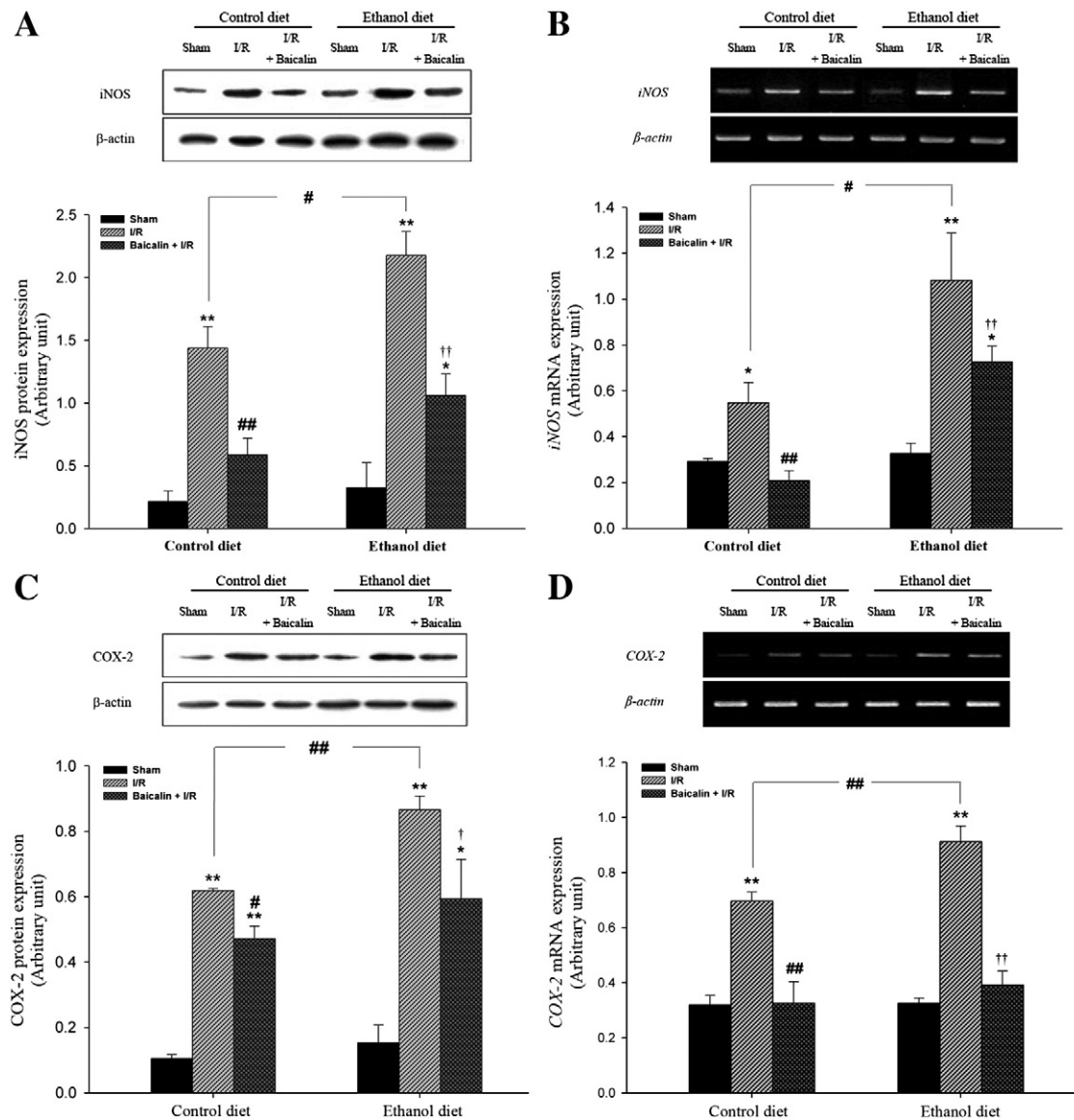


Fig. 5. Effect of baicalin on iNOS, COX-2 protein and mRNA expression after I/R in alcoholic fatty liver. A: hepatic iNOS protein expression. B: hepatic iNOS mRNA expression. C: hepatic COX-2 protein expression. D: hepatic COX-2 mRNA expression. The density values were normalized to β -actin. Values are means $\pm$ SEM for 8–10 rats per group. *,**Significantly different ($P < 0.05$, $P < 0.01$) vs. each sham group. ###Significantly different ($P < 0.05$, $P < 0.01$) vs. CD I/R. †,††Significantly different ($P < 0.05$, $P < 0.01$) vs. ED I/R.

significantly increased after reperfusion, and the two mediators were markedly increased in animals subjected to chronic alcohol consumption, suggesting that chronic alcohol consumption primes the host for an enhanced inflammatory response. Baicalin attenuated the increases in iNOS and COX-2 protein and mRNA expression.

In conclusion, our findings suggest that alcoholic fatty livers augmented I/R-induced inflammatory responses, which may be related to primary graft nonfunction of marginal livers. Moreover, these findings document that baicalin offers protection against posts ischemic injury in alcoholic fatty liver by reducing the hepatocellular damage through suppression of the TLR4-mediated inflammatory signaling pathway. Therefore, pretreatment with baicalin might be useful as a potential pharmacological maneuver for attenuating I/R injury in alcoholic fatty liver.

Conflicts of interest

The authors have no potential financial conflicts of interest.

Acknowledgments

This research was supported by the Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education, Science and Technology (2010-0028646).

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